

Kanan Mahammadli

PhD Student, Max Planck Institute for Informatics — CS@Max Planck PhD Program

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PROFILE

First-year PhD student in the CS@Max Planck program with a background in mathematics, statistics, representation learning, optimization, and generative modelling. My current research studies how learned representations support sequential decision-making and why model performance deteriorates in long-horizon and compositional tasks. Broadly interested in foundation-model adaptation, reasoning and generalization, rigorous evaluation, and methods for understanding and improving model behaviour.

RESEARCH INTERESTS

Foundation Models, Model Adaptation and Post-Training, Reasoning and Generalization, Model Evaluation, Representation Learning

EDUCATION

Max Planck Institute for Informatics

PhD in Computer Science, CS@Max Planck PhD Program

Saarbrücken, Germany

Sep. 2025 – Present

Middle East Technical University

B.S. in Mathematics, Minor in Statistics — CGPA: 3.64/4.00

Ankara, Turkey

Sep. 2020 – Jun. 2025

- Graduated with High Honor / Honor distinction; recipient of Full Tuition Scholarship for Academic Excellence.
- Relevant graduate coursework: Machine Learning, Deep Learning, Reinforcement Learning, Sequence Models in Multimedia.

RESEARCH EXPERIENCE

PhD Student Researcher

Max Planck Institute for Informatics

Oct. 2025 – Jul. 2026

Saarbrücken, Germany

- Completed two Research Immersion Labs, supervised by Prof. Jan Eric Lenssen and Dr. Marc Habermann, on self-supervised latent-action representations for robot learning.
- Investigated continuous action representations learned from human videos and their use as task-relevant conditioning for downstream robot action generation.
- Explored whether learned visual-transition representations capture information needed for sequential decision-making, combining representation learning with diffusion- and flow-matching-based action generation.
- The project motivated broader research questions on representation sufficiency, long-horizon decision-making, compositional generalization, and interpretable failure analysis in learned autonomous systems.

Undergraduate Student Researcher

AI Lab, METU BILTIR Center — Supervised by Prof. Seyda Ertekin

Jul. 2023 – Jun. 2025

Ankara, Turkey

- Conducted research on automated and efficient hyperparameter optimization using large language models, leading to the SLLMBO framework.

Undergraduate Computer Vision Research Intern

University of Houston — Supervised by Prof. Ioannis Kakadiaris

Jul. 2022 – Sep. 2022

Houston, TX, USA

- Worked on human detection from UAV imagery captured under varying viewpoints, distances, and weather conditions, using feature pyramid networks for multi-scale visual recognition.
- Worked on face representation and clustering using FaceNet embeddings and FAISS for efficient similarity-based analysis.

PREPRINTS

Mahammadli, K. and Ertekin, S. *Sequential Large Language Model-Based Hyperparameter Optimization*. arXiv:2410.20302, 2024.

SELECTED INDUSTRY EXPERIENCE

Senior Data and Optimization Scientist

Nov. 2023 – Jul. 2025

SmartKiwi AI

London, UK (Remote)

- Built an end-to-end decision-support platform for demand forecasting and inventory optimization, enabling users to obtain tailored forecasts and support operational decisions through an interactive interface.
- Developed and deployed a high-frequency passenger forecasting and optimization system at Istanbul Grand Airport for lane and staff allocation, achieving 92% accuracy, reducing wait times by 25–35%, and improving staff productivity.

Mid-level Data Scientist

May 2023 – Nov. 2023

SmartKiwi AI

London, UK (Remote)

- Designed a polygon-based region assignment strategy for Sendeo’s courier operations, replacing street-based assignments with optimized service regions and improving delivery coverage and workload balance.
- Contributed to automated courier scheduling and route assignment systems, replacing manual planning with data-driven optimization across the delivery network.

Junior Data Scientist

Jan. 2023 – May 2023

SmartKiwi AI

London, UK (Remote)

- Developed a geospatial demand modeling strategy for car-sharing systems by partitioning cities into hexagonal grids, improving vehicle distribution and service availability across multiple deployments.
- Built a surge pricing model for more than 80 kitchen locations by analyzing user behavior and dynamically adjusting delivery fees, increasing monthly delivery revenue by 50% without harming user conversion.

Data Science Intern

Nov. 2021 – Apr. 2022

Affable AI

Singapore (Remote)

- Improved the sentiment analysis pipeline, reducing batch prediction latency from 180 seconds to 23 seconds and enabling faster large-scale NLP processing.
- Deployed a transformer-based brand classification model that improved influencer matching quality by 25% for digital marketing applications.

TECHNICAL SKILLS

Programming / Tools: Python, PyTorch, scikit-learn, NumPy, Pandas, Git

Areas: Deep Learning, Computer Vision, Representation Learning, Machine Learning, Optimization